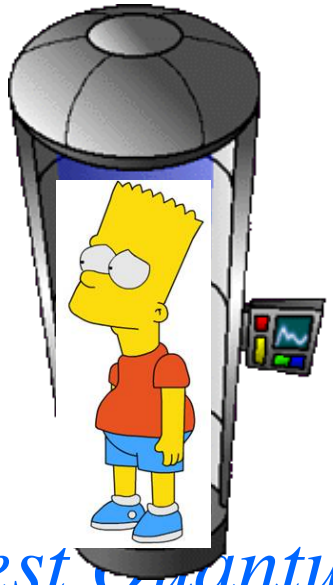
**Nothing To See Here**
**(Move Along)**

# When Is Teleportation Quantum?



**Dave Bacon**

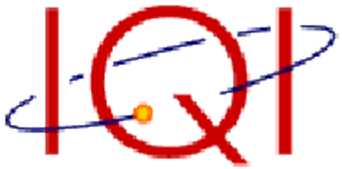
*Department of Physics*

*Caltech*

*Southwest Quantum Information and Technology Research Network (SQITRN)*








# SQI TRN:



# In Collaboration With…

**Ben Toner**

*graduate student*

*Department of Physics*

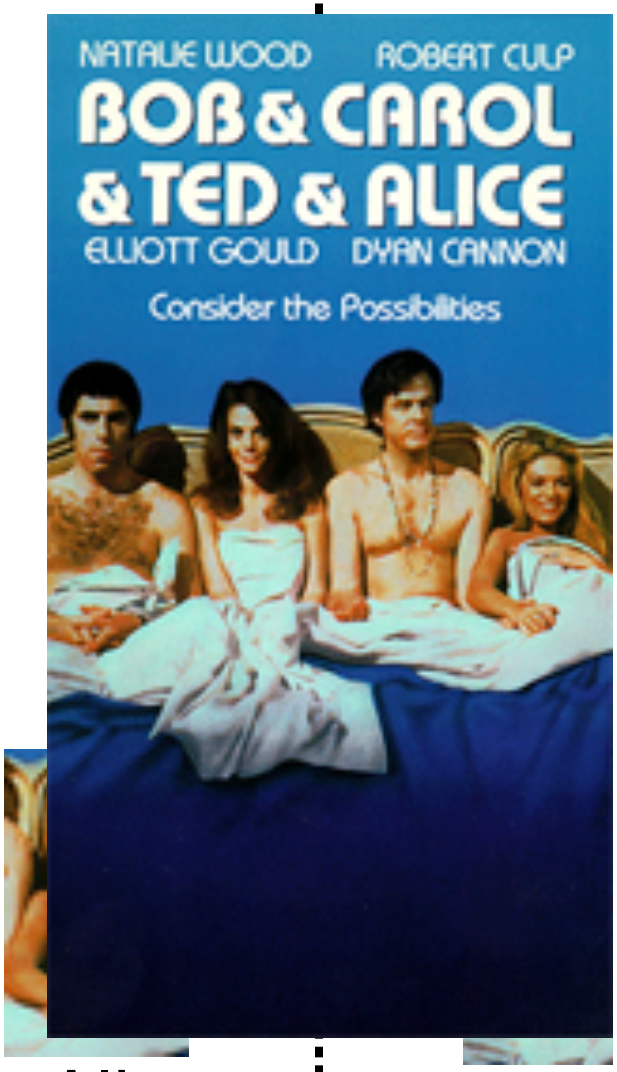
*Institute for Quantum Information*

*Caltech*



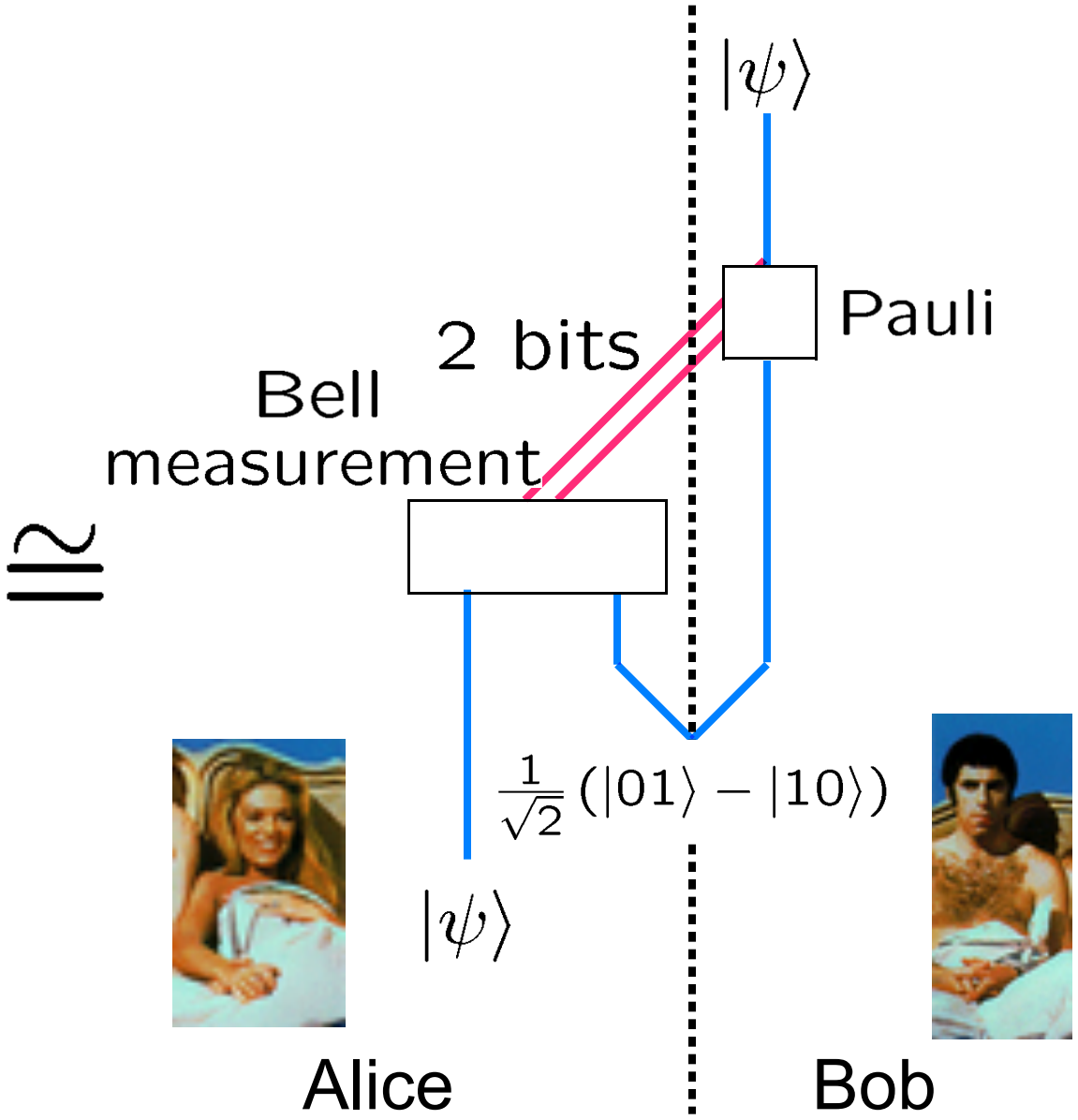
**see Ben's poster**
**for connections to reality**

# Teleportation




Alice Bob

$\cong$




Alice Bob

# An Experiment

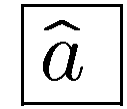
**"qubit teleporation"**



Alice



Bob



$\hat{b}$ measurement $\mathbf{P}_{\pm} = \frac{1}{2}\left(\mathbf{I} \pm \hat{b} \cdot \hat{\boldsymbol{\sigma}}\right)$

Prob(+1: along $\hat{b}$)$=\frac{1}{2}\left(1 + \hat{a} \cdot \hat{b}\right)$

Prob(-1: along -$\hat{b}$)$=\frac{1}{2}\left(1 - \hat{a} \cdot \hat{b}\right)$

2 bits

$\frac{1}{\sqrt{2}}(|01\rangle - |10\rangle)$

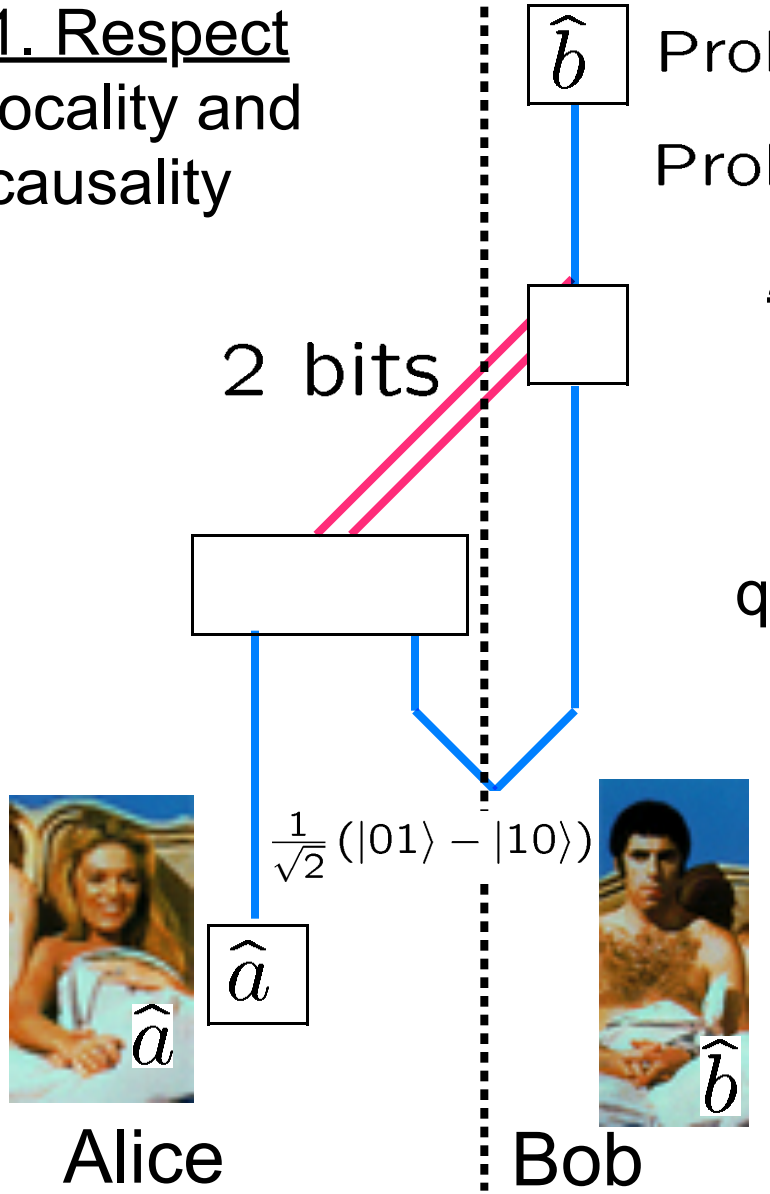
preparation $|\psi\rangle$

$\rho = \frac{1}{2}(\mathbf{I} + \hat{a} \cdot \hat{\boldsymbol{\sigma}})$

In what sense is this experiment "quantum"?

# Simulating Classically

1. Respect locality and causality




$$\text{Prob}(+1\text{: along } \hat{b}) = \frac{1}{2}\left(1 + \hat{a}\cdot\hat{b}\right)$$

$$\text{Prob}(-1\text{: along } -\hat{b}) = \frac{1}{2}\left(1 - \hat{a}\cdot\hat{b}\right)$$

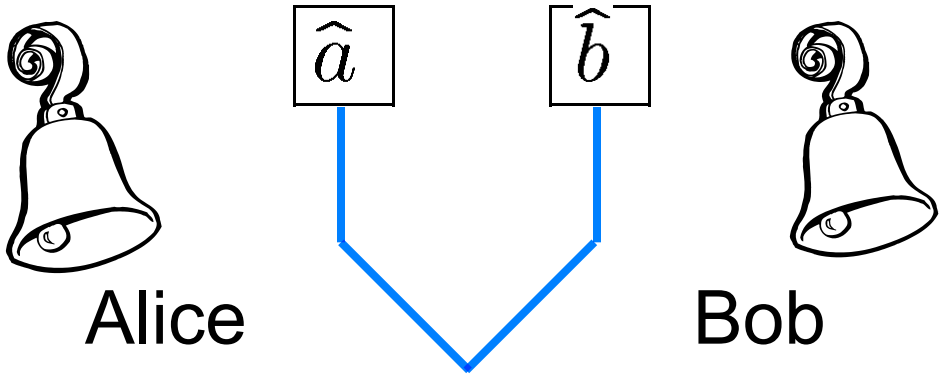
2. replace

quantum wire by classical wire carrying any amount of classical info

classical wire b bits by classical wire b bits

# Simulating Bell Correlations

A's output: $\alpha \in \pm 1$ $\beta \in \pm 1$:Bob's output




$$\frac{1}{\sqrt{2}}(|01\rangle - |10\rangle)$$

$$\Pr(\alpha, \beta | \hat{a}, \hat{b}) = \frac{1}{4}\left(1 - \alpha\beta \hat{a} \cdot \hat{b}\right)$$

simulating Bell Correlations with b bits communication and shared randomness $\Longrightarrow$ simulating qubit teleportation with b+1 bits communication and shared randomness

"Classical teleportation of a qubit" Cerf, Massar, Gisin (2000)

# Bell Pair to Qubit Teleportation

Alice and Bob can produce

$$\Pr(\alpha, \beta | \hat{a}, \hat{b}) = \frac{1}{4}\left(1 - \alpha\beta \hat{a} \cdot \hat{b}\right)$$

Alice sends her output bit (α) to Bob

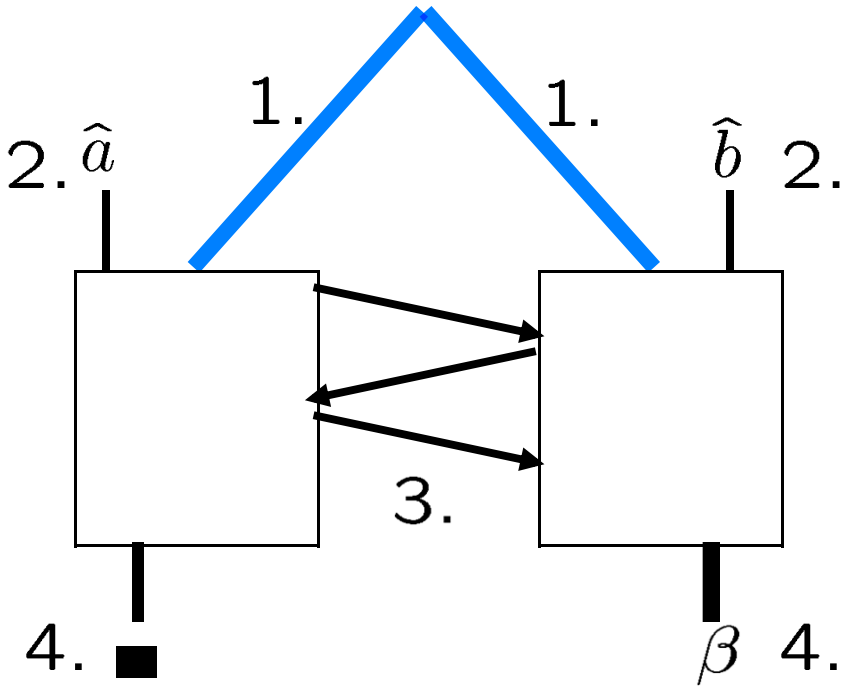
If Alice's output bit equals Bob's output bit,
Bob outputs -1
Otherwise
Bob outputs +1

**qubit teleportation**

$$\Pr(\alpha = \beta | \hat{a}, \hat{b}) = \frac{1}{2}\left(1 - \hat{a} \cdot \hat{b}\right) \quad \Pr(\text{B outputs } -1) = \frac{1}{2}\left(1 - \hat{a} \cdot \hat{b}\right)$$

$$\Pr(\alpha \neq \beta | \hat{a}, \hat{b}) = \frac{1}{2}\left(1 + \hat{a} \cdot \hat{b}\right) \quad \Pr(\text{B outputs } +1) = \frac{1}{2}\left(1 + \hat{a} \cdot \hat{b}\right)$$

# Simulating Bell Correlations



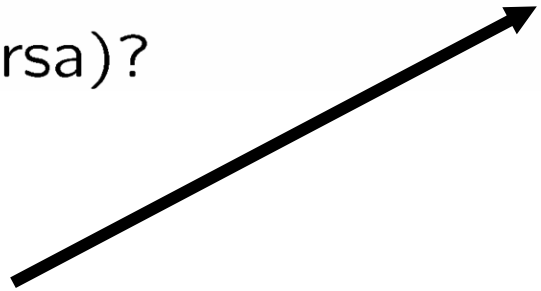

1. distribute shared random bits
2. A and B choose measurement directions, $\hat{a}$, $\hat{b}$
3. A and B communicate $b$ bits
4. A and B output $\alpha, \beta \in \{\pm 1\}$

$$\Pr(\alpha, \beta | \hat{a}, \hat{b}) = \frac{1}{4}\left(1 - \alpha\beta \hat{a} \cdot \hat{b}\right)$$

# Why Is It Hard?

$$\Pr(\alpha, \beta | \hat{a}, \hat{b}) = \frac{1}{4}\left(1 - \alpha\beta \hat{a} \cdot \hat{b}\right)$$

How to produce distribution without sending $\hat{a}$ from A to B (or vice versa)?



requires infinite number of bits

Can be done with 8 bits + shared randomness

Brassard, Cleve, Tapp (1999)

8 bits?

# Towards Simulating Bell

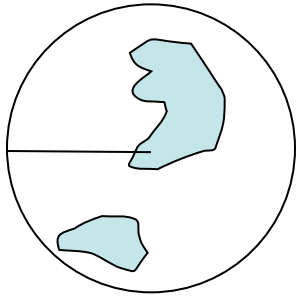
$$\Pr(\alpha, \beta | \hat{a}, \hat{b}) = \frac{1}{4}\left(1 - \alpha\beta \hat{a} \cdot \hat{b}\right)$$

Notice:

1. If $\hat{a} = \hat{b}$, then $Pr(\alpha = \beta) = 0$

   If $\hat{a} = -\hat{b}$, then $Pr(\alpha = \beta) = 1$

If B learns from A, that A's vector lies in some region



$\Downarrow$

forces B's output

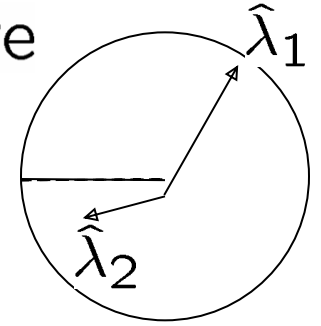
2. $\Pr\left(\alpha, \beta | \hat{a}, \hat{b}\right)$ is symmetric under joint rotations.

$$\Pr\left(\alpha, \beta | \hat{a}, \hat{b}\right) = \Pr\left(\alpha, \beta | R(\hat{a}), R(\hat{b})\right)$$

# A 1 Bit Protocol

Two shared random variables:

$\hat{\lambda}_1, \hat{\lambda}_2$ uniformly distributed on sphere



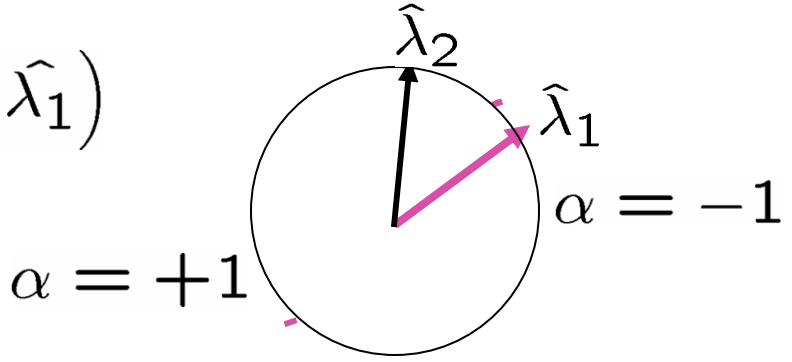

1. Alice outputs $-\text{sgn}\left(\hat{a} \cdot \hat{\lambda_2}\right)$

2. Alice sends a bit to Bob: $c = \text{sgn}\left(\hat{a} \cdot \hat{\lambda}_1\right) \text{sgn}\left(\hat{a} \cdot \lambda_2\right)$

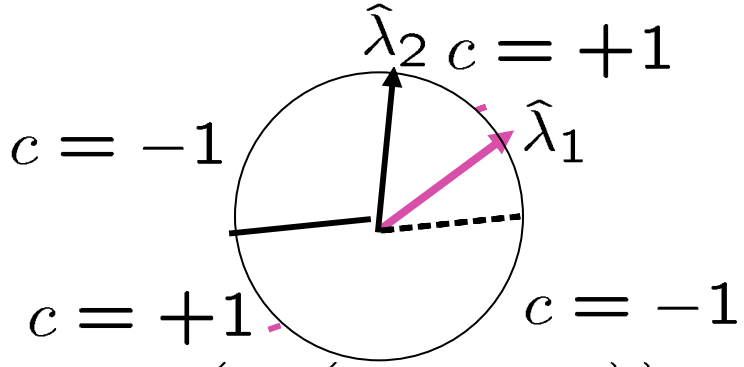
3. Bob outputs $\text{sgn}\left(\hat{b} \cdot \left(\hat{\lambda_1} + c\hat{\lambda_2}\right)\right)$

# The Intuition

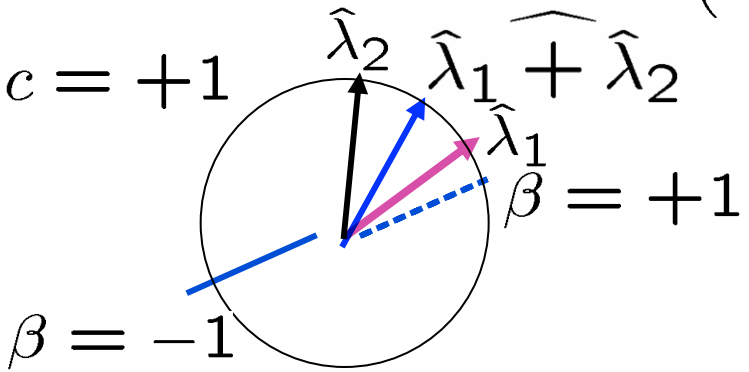
1. Alice outputs $-\text{sgn}\left(\hat{a}\cdot\hat{\lambda_1}\right)$

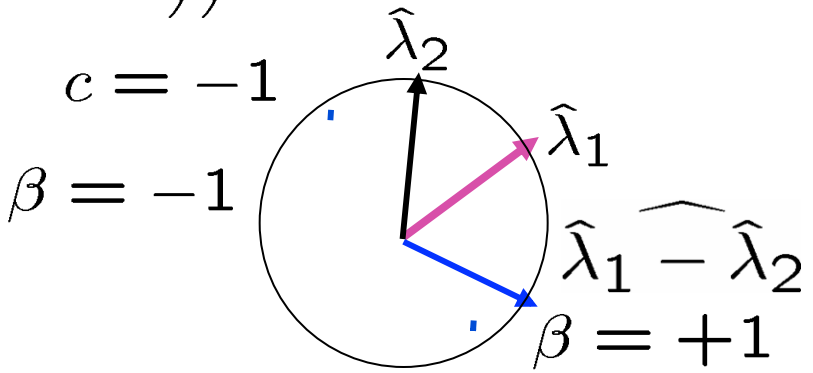



2. Alice sends a bit to Bob: $c=\text{sgn}\left(\hat{a}\cdot\hat{\lambda}_1\right)\text{sgn}\left(\hat{a}\cdot\lambda_2\right)$




3. Bob outputs $\text{sgn}\left(\hat{b}\cdot\left(\hat{\lambda_1}+c\hat{\lambda_2}\right)\right)$



# It Works!

Each parties output changes sign under

$$\hat{\lambda}_1 \to -\hat{\lambda}_1,\ \hat{\lambda}_2 \to -\hat{\lambda}_2 \quad \Rightarrow \langle\alpha\rangle = \langle\beta\rangle = 0$$

$$\langle\alpha\beta\rangle = E\left[-\mathrm{sgn}(\hat{a}\cdot\hat{\lambda}_1)\sum_{d=\pm 1}\frac{1+cd}{2}\,\mathrm{sgn}\left[\hat{b}\cdot(\hat{\lambda}_1+d\hat{\lambda}_2)\right]\right]$$

$$E[x] = \frac{1}{(4\pi)^2}\int d\hat{\lambda}_1\int d\hat{\lambda}_2[x] \qquad c = \mathrm{sgn}\left(\hat{a}\cdot\hat{\lambda}_1\right)\mathrm{sgn}\left(\hat{a}\cdot\lambda_2\right)$$

$$\langle\alpha\beta\rangle = E\left[-2\,\mathrm{sgn}(\hat{a}\cdot\hat{\lambda}_1)\,\mathrm{sgn}\left[\hat{b}\cdot(\hat{\lambda}_1+\hat{\lambda}_2)\right]\right]$$

$$\frac{1}{4\pi}\int d\hat{\lambda}_2 \mathrm{sgn}\left(r+\hat{b}\cdot\hat{\lambda}_2\right) = r$$

$$-\frac{1}{4\pi}\int d\hat{\lambda}_1 2\mathrm{sgn}\left(\hat{a}\cdot\hat{\lambda}_1\right)\hat{b}\cdot\hat{\lambda}_1 = -\hat{a}\cdot\hat{b}$$

$$\langle\alpha\beta\rangle = -\hat{a}\cdot\hat{b}$$

# 1 Bell Pair's Worth of Correlations

correlations from projective measurements on Bell pair

$=$

shared randomness

$+$

1 bit of communication

# Qubit Teleportation

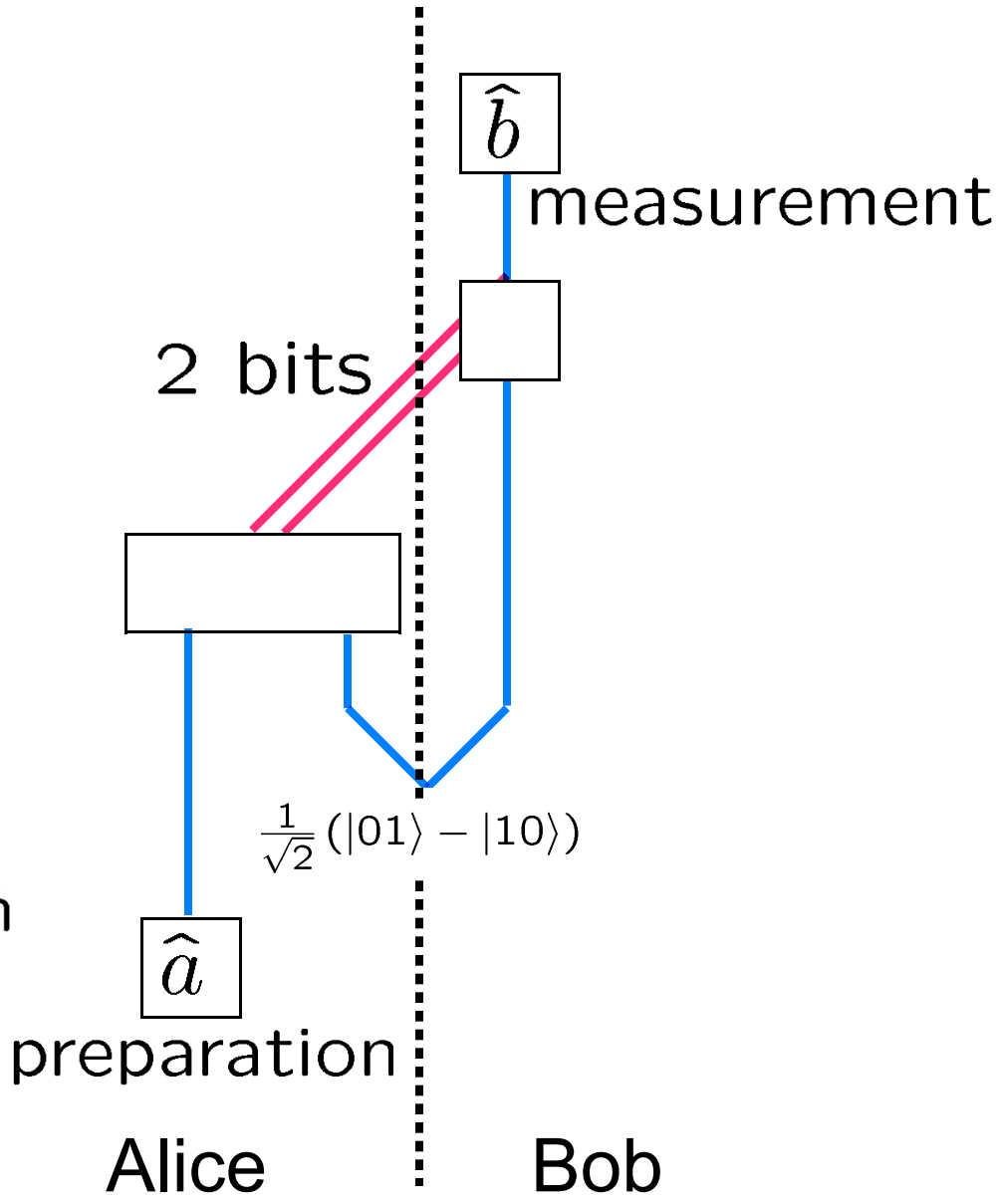
qubit teleportation

=

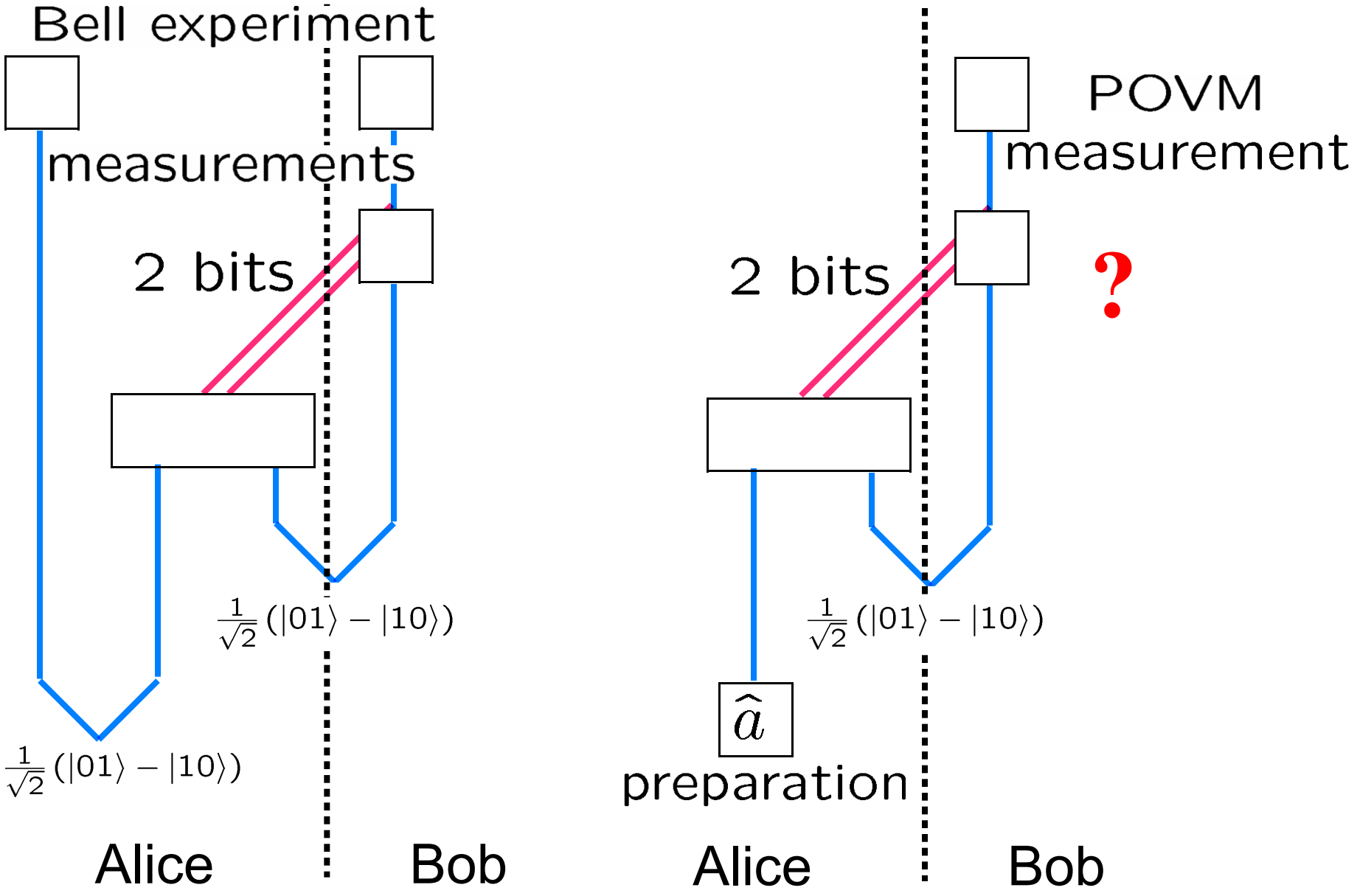
shared randomness

+

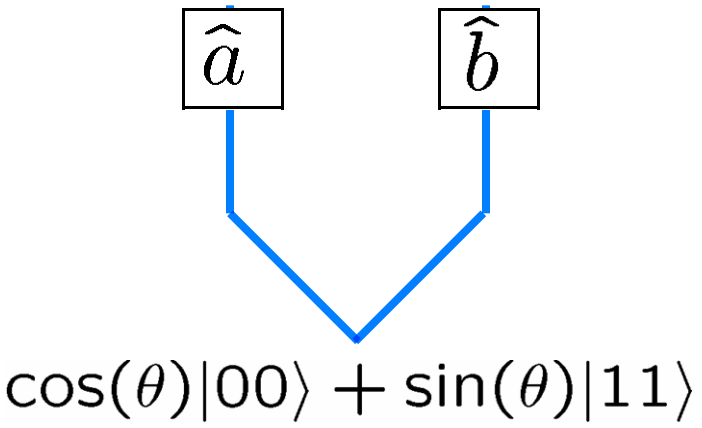
2 bits of communication



# Quantum Teleportation



# Correlations on Non-maximally Entangled 2 Qubit States




Alice simulates measurement along $\hat{a}$, output $\alpha$

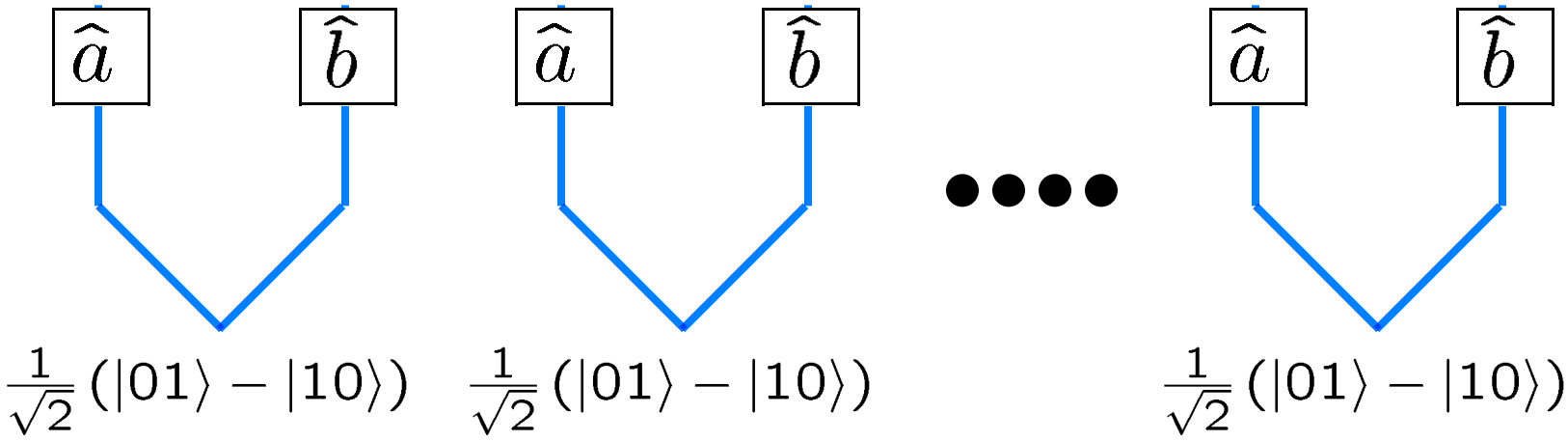
Alice now can calculate what state Bob should have:

Call state $\hat{a}'$

Parties run qubit teleportation scheme to make Bob measure $\hat{a}'$ along $\hat{b}$, output $\beta$

2 bits

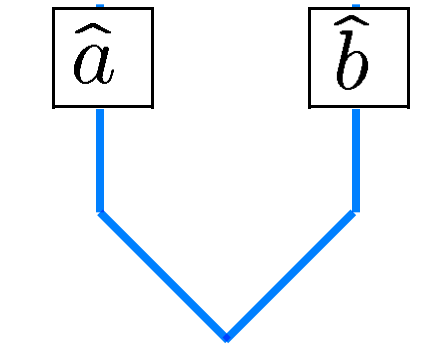
# Average Communication




Simulate the correlations for $n$ Bell pair experiments.

Can compress amount of communication to $\approx 0.85n$ bits.

# Connection to Detector Inefficiency




1. Alice outputs
2. Alice sends bit to Bob
3. Bob outputs

Distribute extra random bit, 50% 0, 50% 1

A and B run protocol as if random bit was communicated bit

If A finds that she was going to send a bit $\neq$ random bit,

Alice says: "my detector didn't fire"

If A finds that she was going to send a bit $=$ random bit,

Great. Protocol has produced correlations

Symmetrize + remove detector correlations

Ben Toner's poster



local hidden variable model with detector inefficiency

# Futurama

Simulating correlations on general states with general measurements (communication bounded by size of Hilbert space)

Why does this work?

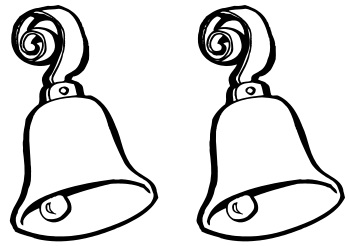
How do we justify connection between classical protocols and Hilbert space probabilities?

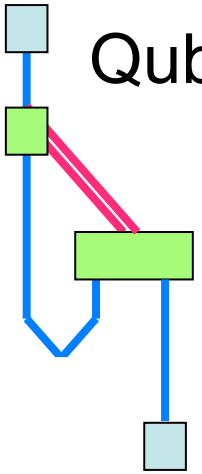
What is the two-party game with communication from which we can derive quantum theory?

Cryptographic interpretation of quantum theory

# Fin

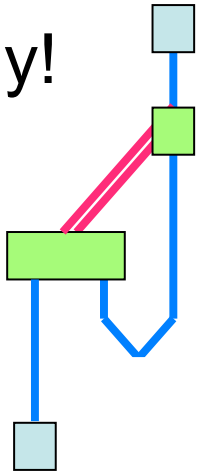
Correlations on Bell Pair = shared randomness + 1 bit





Qubit teleportation can be simulated classically!



see Ben Toner's poster for connections to the detector inefficiency loophole for Bell inequalities